Crouzeix: Folie à Deux

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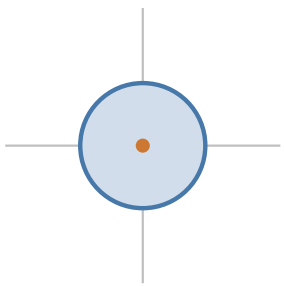
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Numerical range

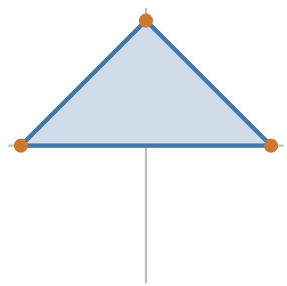
Definition. Let $\mathcal{H}$ be a complex Hilbert space and $T \in \mathcal{B}(\mathcal{H})$ a bounded operator. The numerical range of T is

$$W(T) = \{\langle Tx, x\rangle : \ x \in \mathcal{H}, \ \|x\| = 1\}.$$

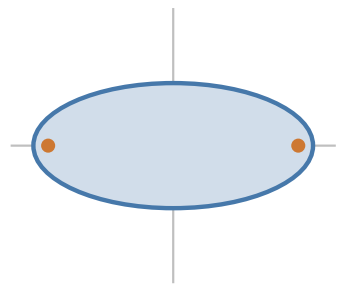
In finite dimension $W(T)$ is compact and convex (Toeplitz–Hausdorff); in general it is convex, and $\overline{W(T)} \supseteq \operatorname{conv} \sigma(T)$.



$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$



$\operatorname{diag}(1, i, -1)$



$\begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}$

Orange: spectrum. Blue: numerical range. A normal operator has $\overline{W(T)} = \operatorname{conv} \sigma(T)$; a 2×2 matrix has an ellipse with foci at the eigenvalues.

Spectral sets

Definition. A compact set $S \subset \mathbb{C}$ is a *K-spectral set* for $T \in \mathcal{B}(\mathcal{H})$ if

$$\|p(T)\| \leq K \sup_{z \in S} |p(z)| \qquad \text{for all } p \in \mathbb{C}[z].$$

(For convex S, functions holomorphic near S come along by Runge.)

von Neumann's inequality (1951). The closed disk is a spectral set for every contraction:

$$\|T\| \leq 1 \;\Rightarrow\; \|p(T)\| \leq \sup_{\mathbb{D}} |p|.$$

Also the closed right half-plane, when $\operatorname{Re} T \geq 0$.

Two proofs:

- **Function theory.** von Neumann: reduce to Blaschke products, check Möbius maps by hand.
- **Dilation theory.** Sz.-Nagy: T is the compression of a unitary U, and $p(T) = P_{\mathcal{H}}\, p(U)|_{\mathcal{H}}$.

Crouzeix's conjecture

Conjecture (Crouzeix 2004)

Let T be a bounded operator on a Hilbert space $\mathcal{H}$. Then $\overline{W(T)}$ is a 2-spectral set for T.

Remark. The constant is sharp. For $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $p(z) = z$,

$$W(N) = D(0, \tfrac{1}{2}), \qquad \|p(N)\| = 1 = 2 \cdot \tfrac{1}{2} = 2 \sup_{D(0,1/2)} |p|.$$

Antecedents.

- von Neumann: constant 1, only for contractions.
- Berger 1965, Berger–Stampfli 1967, Okubo–Ando 1975: the disk is a 2-spectral set whenever it contains $W(T)$. Via 2-dilations: $T^n = 2V^*U^nV$.

So Crouzeix asks for the disk's constant on every convex set. For 2×2 matrices he proved it, with equality only when $W(T)$ is a disk.

Constants

Ω bounded convex with smooth boundary, $W(T) \subset \Omega$; $A(\Omega)$ the functions holomorphic on Ω, continuous on $\overline{\Omega}$.

$$C_\Omega := \sup\left\{\|f(T)\| : \ W(T) \subset \Omega, \ \|f\|_\infty \le 1\right\}.$$

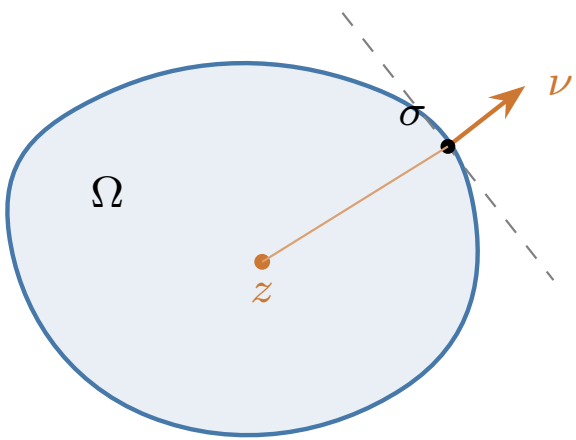
Delyon–Delyon	1999	$C_\Omega < \infty$	depends on Ω
Crouzeix	2004	2	only for 2×2
Crouzeix	2007	11.08	uniform in Ω
Crouzeix–Palencia	2017	$1 + \sqrt{2}$	completely bounded
Jin	27 July 2026	2	function theory
Lorist–Schwenninger	Aug 2026	2	dilation theory

Technique, 1999 to 2026: one object, the double-layer potential. Shrinking Ω to $\overline{W(T)}$, $\sup_\Omega C_\Omega \le 2$ is the conjecture.

Double-layer potential

For $\sigma \in \partial\Omega$ let $\nu = \nu(\sigma)$ be the outward unit normal and ds arclength. For $z \in \Omega$,

$$\mu(\sigma, z) = \frac{1}{2\pi}\Big(\underbrace{\frac{\nu}{\sigma - z}}_{\text{first layer}} + \underbrace{\frac{\bar{\nu}}{\bar{\sigma} - \bar{z}}}_{\text{second layer}} \Big) = \frac{1}{\pi} \operatorname{Re} \frac{\nu}{\sigma - z}.$$



- First layer: the Cauchy kernel, $\frac{1}{2\pi}\frac{\nu\, ds}{\sigma - z} = \frac{1}{2\pi i}\frac{d\sigma}{\sigma - z}$.
- $\mu\, ds = \frac{1}{\pi}\, d \arg(\sigma - z)$: the angle at which z sees the boundary. So $\mu \geq 0$ by convexity, and $\int \mu\, ds = 2$ for $z \in \Omega$, 1 for $z \in \partial\Omega$.
- Why use it? The first layer gives $f(z)$. If we can control the second layer, we win.

Double-layer potential, continued

Replace z by T with $W(T) \subset \Omega$. For $f \in C(\partial\Omega)$ put

$$C(f,T) = \frac{1}{2\pi i}\int_{\partial\Omega} f(\sigma)(\sigma - T)^{-1}\,d\sigma,$$

$$S(f,T) = C(f,T) + C(\bar{f},T)^*.$$

S is the functional calculus integrated against the double-layer potential $\mu(\sigma,T) = \frac{1}{\pi}\operatorname{Re}[\nu(\sigma)(\sigma - T)^{-1}]$:

$$S(f,T) = \int_{\partial\Omega} f(\sigma)\,\mu(\sigma,T)\,ds, \qquad \mu(\sigma,T) \geq 0, \qquad \int_{\partial\Omega} \mu\,ds = 2I.$$

Lemma (Crouzeix–Palencia 2017)

For $f \in A(\Omega)$, $\|S(f,T)\| = \|f(T) + C(\bar{f},T)^*\| \leq 2\|f\|_{\partial\Omega}$.

Positivity and total mass $2I$, then Cauchy–Schwarz.

So the question: how does $\|S(f,T)\|$ control $\|f(T)\|$? Can we estimate $C(\bar{f},T)^*$, or make it vanish altogether?

Positivity and the double-layer formula

Write $\Phi(f) = \frac{1}{2} S(f, T)$, a unital positive map $C(\partial\Omega) \to \mathcal{B}(\mathcal{H})$. For $f \in A(\Omega)$,

$$2\Phi(f) = f(T) + \alpha(f)(T)^*, \qquad \alpha(f)(z) = \frac{1}{2\pi i} \int_{\partial\Omega} \frac{\overline{f(\sigma)}\, d\sigma}{\sigma - z},$$

where $\alpha : A(\Omega) \to A(\Omega)$ is antilinear and $\alpha(1) = 1$. Positivity of Φ is the lemma: $\|2\Phi(f)\| \le 2\|f\|$.

For a point z this is $2\Phi(f)(z) = f(z) + \overline{\alpha(f)(z)}$: the weighted average of f on the boundary, split into a holomorphic and an anti-holomorphic piece. For the disk, $\alpha(f) = \overline{f(0)}$, a constant.

Two ways to use the formula.

- Estimate the companion: Crouzeix–Palencia, $\|\alpha(f)\| \le \|f\|$, hence $1 + \sqrt{2}$.
- Study the extremal f, a Blaschke product, and show $\|f(T)\| \le \|2\Phi(f)\|$ for it: Bickel, Gorkin, Greenbaum, Overton. Not the route either 2026 proof takes.

Crouzeix–Palencia: $1+\sqrt{2}$, and why not 2

Let $K = C_\Omega$, $\|f\| \le 1$, $T = f(A)$, $g = \alpha(f)$. Since $\|g\| \le \|f\|$ on convex Ω,

$$TT^*TT^* = T\left(T + g(A)^*\right)^* TT^* - (fgf)(A)\,T^*,$$

so $K^4 \le 2K^3 + K^2$ and $K \le 1+\sqrt{2}$. (Ransford–Schwenninger's four lines)

Only two facts about the companion were used: $\|f(A) + g(A)^*\| \le 2$ and $\|g\| \le \|f\|$. Under those two alone, $1+\sqrt{2}$ is attained: $T = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$, Ω two small disks about 0 and 1, $g = -f(0)$ on one, $-f(1)$ on the other, $h = \mp 1$ gives $\|h(T)\| = 1+\sqrt{2}$.

Not a counterexample, since $W(T) \not\subset \Omega$, but the end of that road. This g sends $1 \mapsto -1$; the Cauchy transform sends $1 \mapsto 1$.

Equivalent formulation

Theorem (Twisted real part)

Let $\mathcal{A}$ be a unital commutative operator algebra, $\theta : \mathcal{A} \to \mathcal{B}(\mathcal{H})$ a unital bounded homomorphism, and $\alpha : \mathcal{A} \to \mathcal{A}$ a unital antilinear map such that the unital linear map

$$\theta_\alpha : \mathcal{A} \to \mathcal{B}(\mathcal{H}), \qquad a \mapsto \tfrac{1}{2}(\theta(a) + \theta(\alpha(a))^*)$$

is contractive. Then $\|\theta\| \leq 2$.

- Ransford–Schwenninger (2018) and Clouâtre–Ostermann–Ransford (2023): this statement implies Crouzeix. Take $\theta(f) = f(T)$, α the Cauchy transform; then $\theta_\alpha = \Phi$.
- Michael Hartz (31 July 2026) proves the theorem, by reorganising Jin's solution.

Unital and contractive implies positive: $\operatorname{Re} a \geq 0 \Rightarrow \operatorname{Re} \theta_\alpha(a) \geq 0$. The hypothesis is positivity of the twisted real part.

Un

Function theory

Jin, 27 July 2026. Operator form: Hartz, unpublished notes, 31 July 2026.

Holomorphic $\to$ resolvent transfer

Theorem (Holomorphic $\to$ resolvent transfer; Jin, operator form Hartz)

Let $T \in \mathcal{B}(\mathcal{H})$ with $\sigma(T) \subset \overline{\mathbb{D}}$. Let $H : \mathbb{D} \to \mathcal{B}(\mathcal{H})$ be holomorphic with $H(0) = I$ and $\operatorname{Re} H(z) \geq 0$ for all $z \in \mathbb{D}$. Assume that

$$\Delta(z) := H(z) - (I - zT)^{-1}$$

commutes with T^* for all $z \in \mathbb{D}$. Then $\|T\| \leq 2$.

- $\Delta = 0$: $\operatorname{Re}(I - zT)^{-1} \geq 0$ is Berger's condition $W(T) \subset \overline{\mathbb{D}}$, and the conclusion is Okubo–Ando. A commuting correction costs nothing.
- Transfer $\Rightarrow$ twisted real part (Hartz). $T = \theta(a)$, $c_z = (1 + za)(1 - za)^{-1}$, $H(z) = \theta_\alpha(c_z)$. Then $\operatorname{Re} H \geq 0$ and $\Delta(z) = \frac{1}{2}(\theta(\alpha(c_z))^* - I)$, an adjoint from the commutative algebra $\theta(\mathcal{A}) \ni T$. Concretely $\Delta(w) = \sum_{m \geq 1} w^m \alpha(f^m)(T)^*$: every power of f at once.

Proof of the transfer theorem

On $H^2 \otimes \mathcal{H}$ (after scaling $T \to rT$), $\operatorname{Re} M_H \geq 0$ and $M_H = M_R + M_\Delta$, $R(z) = (I - zT)^{-1}$.

For W with $\sigma(W) \subset \mathbb{D}$ let $N_W = \{F = \sum F_n z^n : \sum W^n F_n = 0\}$, the functions vanishing at W.

- $\Delta(z)$ commutes with $W \Rightarrow M_\Delta N_W \subset N_W$ (an ideal, as in one variable).
- $\Delta(0) = 0 \Rightarrow$ the kernel Γ of N_W at 0 has $\Gamma^* M_\Delta \Gamma = 0$.

Take $W = \frac{1}{2}T^*$: $\quad \Gamma^*(M_R + M_R^*)\Gamma = \Gamma^*(M_H + M_H^*)\Gamma \geq 0$.

Unwinding, with $P = \sum 4^{-n} T^{*n} T^n$, $Q = \sum 2^{-n} T^{*n} T^n$, $Y = Q - P \geq \frac{1}{4} T^* T$:

$$4Y - YP - PY \geq 0 \;\Rightarrow\; P \leq 2I \;\Rightarrow\; T^*T \leq T^*PT = 4(P - I) \leq 4I.$$

Jin's coordinates: sample the Herglotz kernel of H at $\lambda_i/2$ and at 0, the origin vector chosen to cancel the diagonal correction. Same compression.

Deux

Dilation theory

Lorist and Schwenninger, arXiv 2608.03841, August 2026.

Dilation theory

Lemma (Lorist–Schwenninger)

Let $\mathcal{H}$ be a Hilbert space and $T \in \mathcal{B}(\mathcal{H})$. Suppose there exist a Hilbert space $\mathcal{K}$, a contraction $Q \in \mathcal{B}(\mathcal{K})$ and an isometry $V \in \mathcal{B}(\mathcal{H}, \mathcal{K})$ such that the operators

$$E_n := 2V^* Q^{*n} V - T^{*n}, \qquad n \in \mathbb{N},$$

are uniformly bounded and commute with T. Then $\|T\| \leq 2$.

$E_n = 0$: T^* has a 2-dilation and $\|T\| = \|2V^*QV\| \leq 2$ at once. The lemma tolerates an error of any size, if bounded in n and commuting with T.

The dilation is Stinespring for Φ: $\mathcal{K} = L^2(\partial\Omega, ds; \mathcal{H})$, $Vx = \frac{1}{\sqrt{2}}\mu(\cdot, T)^{1/2}x$, $Q = M_f$ with $\|f\|_\infty = 1$. Then $V^*Q^nV = \Phi(f^n)$, and with $T = f(A)$, $E_n = 2\Phi(f^n)^* - f(A)^{*n} = \alpha(f^n)(A)$, which commutes with $f(A)$ and is bounded by $\|\theta\|\|\alpha\|$. Hence $\|f(A)\| \leq 2$.

Proof of the lemma

$\kappa := \|T\| > 1$ and x a unit vector with $T^*Tx = \kappa^2 x$ (ultrapower if necessary). Put

$$m_n := \mathrm{Re}\langle E_n T^n x, x\rangle, \qquad u := Q^*VTx - \kappa V x.$$

Two estimates on the same error vector:

$\|u\|^2 \le 2\kappa^2 - \kappa^3 - \kappa m_1$ (V isometric, Q contractive, $n = 1$ only)

$\kappa m_1 \ge -\dfrac{\|u\|^2}{(\kappa-1)^2}$ (telescope $\kappa m_n - m_{n+1}$ over all n)

Together, $\|u\|^2(1-(\kappa-1)^{-2}) \le \kappa^2(2-\kappa)$: impossible if $\kappa > 2$.

The first line alone gives $\kappa \le 1 + \sqrt{1-m_1}$; with $\|\alpha\| \le 1$, $m_1 \ge -1$, and that is $1+\sqrt{2}$ again. $m_1 < 0$ happens for some 3×3 matrices. The second line needs E_n for every n.

The same proof twice

	Un (Jin, Hartz)	Deux (Lorist–Schw.)
positivity	$\operatorname{Re} H \geq 0$	$\Phi = V^* \pi(\cdot) V$
all powers	$H(w) = I + 2\sum_m w^m \Phi(f^m)$	E_n, every n
companion	$\Delta(w) = \sum_m w^m \alpha(f^m)(T)^*$	$E_n = \alpha(f^n)(T)$
used	commutes with T^*	commutes with T
removal	$\Gamma^* M_\Delta \Gamma = 0$	telescoping sum

Crouzeix–Palencia use the *size* of the companion and get $1 + \sqrt{2}$, which is sharp for that information. The new proofs use its *location*: the commutative algebra $\{g(T)\}$ containing $f(T)$. Its size is never used.

The canary

What the papers say the model did.

- Jin: ChatGPT proposed sampling the Herglotz kernel at $\lambda_i/2$ and adding the origin sample that cancels the correction. That is the compression in the transfer proof.
- Lorist–Schwenninger: GPT-5.6, 25–26 July, asked for an improvement in estimates involving the iterates f^n, pointed out that $\Phi(f^{n+1})^* f(A)x - \|f(A)\|\Phi(f^n)^* x$ is controlled uniformly in n by the quantity behind m_1. The lemma was abstracted from that. They saw Jin's preprint on 28 July.
- Hartz: ChatGPT to discuss Jin's preprint.

The credited step is the same in both cases: stop estimating the companion of a single f, and use the identity for all f^n simultaneously. The proofs were written and checked by the authors.

The completely bounded conjecture

Conjecture

For every $m \geq 1$ and every $F = [f_{ij}] \in M_m(A(\Omega))$, $\|F(T)\| \leq 2 \sup_{z \in \Omega} \|F(z)\|$.

Equivalently $f \mapsto f(T)$ has cb-norm ≤ 2; by Paulsen, T is then similar to an operator having $\overline{W(T)}$ as a complete spectral set.

Known: $1 + \sqrt{2}$ (Crouzeix–Palencia); 2 for disks, and for weighted shifts (Crouzeix–Greenbaum 2025).

Neither Un nor Deux amplifies: $M_m(A(\Omega))$ is not commutative, and both proofs rest on the companion commuting with $f(T)$. Crouzeix–Palencia rest only on $\|\alpha\| \leq 1$, which survives.

As of last week: [reference], from the OpenAI SlopDrop (Burnstein, Iosevich, Krause, Yavicoli).

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Appendix: the computation behind the transfer proof

$K_W := E_W^*$, so $(K_W x)(z) = (I - zW^*)^{-1}x$ and $K_0 x \equiv x$. With $W = \frac{1}{2}T^*$: $P = K_W^* K_W \geq I$, and $K_W P^{-1} K_W^*$ is the projection onto $\operatorname{ran} K_W = N_W^\perp$. Hence

$$\Gamma = (I - K_W P^{-1} K_W^*) K_0 = K_0 - K_W P^{-1} \qquad (K_W^* K_0 = I).$$

The resolvent identity
$(I - zT)^{-1}(I - zT/2)^{-1} = 2(I - zT)^{-1} - (I - zT/2)^{-1}$ gives

$$M_R K_W = 2K_{T^*} - K_W, \qquad M_R K_0 = K_{T^*}, \qquad M_R^* K_0 = K_0.$$

With $Q = K_W^* K_{T^*}$,

$$\begin{pmatrix} K_W^* \\ K_0^* \end{pmatrix} (M_R + M_R^*) \begin{pmatrix} K_W & K_0 \end{pmatrix} = \begin{pmatrix} 4Q - 2P & Q + I \\ Q + I & 2I \end{pmatrix},$$

and therefore $\Gamma^*(M_R + M_R^*)\Gamma = P^{-1}(4Y - YP - PY)P^{-1}$ with $Y = Q - P$.

Appendix: from $4Y - YP - PY \geq 0$ to $\|T\| \leq 2$

$P = \sum_{n \geq 0} 4^{-n} T^{*n} T^n \geq I$ and $Y \geq \frac{1}{4} T^* T \geq 0$.

Suppose $Px = \alpha x$ with $\|x\| = 1$ and $\alpha > 2$. Then

$$0 \leq \langle (4Y - YP - PY)x, x \rangle = (4 - 2\alpha)\langle Yx, x \rangle,$$

so $\langle Yx, x \rangle = 0$, so $Yx = 0$, so $Tx = 0$, so $Px = x$: a contradiction. In general use approximate eigenvectors, or an ultrapower. Hence $I \leq P \leq 2I$.

The series defining P satisfies $P = I + \frac{1}{4} T^* P T$, so

$$T^* T \leq T^* P T = 4(P - I) \leq 4I.$$

Appendix: the two estimates in the proof of the lemma

Since E_n commutes with T and $T^*Tx = \kappa^2 x$,

$$\begin{aligned}
\kappa m_n - m_{n+1} &= \operatorname{Re}\langle T^n(\kappa E_n - E_{n+1}T)x, x\rangle \\
&= \kappa(\kappa-1)\|T^{*n}x\|^2 - 2\operatorname{Re}\langle V^*Q^{*n}u, T^{*n}x\rangle \\
&= \kappa(\kappa-1)\left\|T^{*n}x - \frac{V^*Q^{*n}u}{\kappa(\kappa-1)}\right\|^2 - \frac{\|V^*Q^{*n}u\|^2}{\kappa(\kappa-1)} \\
&\geq -\frac{\|u\|^2}{\kappa(\kappa-1)}.
\end{aligned}$$

Multiply by κ^{-n+1} and sum over $n \geq 1$; the left side telescopes to κm_1 because (m_n) is bounded.

For the upper bound,
$\|u\|^2 = \|Q^*VTx\|^2 + \kappa^2 - 2\kappa \operatorname{Re}\langle V^*Q^*VTx, x\rangle$, with
$\|Q^*VTx\| \leq \kappa$ and $2V^*Q^*V = T^* + E_1$:

$$\|u\|^2 \leq 2\kappa^2 - \kappa \operatorname{Re}\langle T^*Tx, x\rangle - \kappa m_1 = 2\kappa^2 - \kappa^3 - \kappa m_1.$$